Yuvraj Kashyap

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Full-Stack Software Engineer

EDUCATION

University of Texas at Dallas Expected May 2027
B.S. in Computer Science, Minor in Finance, Certificate in Entrepreneurship & Innovation
GPA: 3.77/4.00 | Dean's List | NCAA Division II Tennis
Relevant Coursework: Advanced Algorithms, Data Structures & Algorithms, Database Systems, Systems Programming, Software Engineering, Programming Languages, Machine Learning, Artificial Intelligence, Computer Vision, Linear Algebra, Probability & Statistics

EXPERIENCE

Base Power Incoming
Software Engineering Intern *Austin, TX*

Medceptor – HealthTech startup Apr 2026 – Aug 2026
Software Engineering Intern
- Enabled CNA/LPN market expansion by shipping question-bank and exam workflows for a platform serving 1,800+ paying users; owned technical decisions and delivery from vague requirements through production on a five-person engineering team.
- Cut import and validation time per 100-question batch from 70 to 15 minutes (about 79%) by automating manual work with Next.js, TypeScript, Django, and Supabase/Postgres; built track/level filtering, scoring, and review workflows.

UT Dallas, UAV / Smart City Systems Oct 2025 – Present
Undergraduate Researcher
- Reduced required observer positions from 11 to six (45%) while maintaining 93.3% coverage of 240 simulated targets, using Python line-of-sight analysis and greedy set cover across 1,553 Dallas building footprints.
- Built a reproducible UAV simulation framework for a 4 km × 4 km urban environment, implementing altitude-aware A* planning to generate a 4.99 km obstacle-avoiding route.

NOVA Autonomous Driving, UT Dallas Sep 2023 – Aug 2025
Software Research Assistant
- Reduced 3D sensor-data processing time with Python voxel-grid downsampling, cutting scan size by 94.5% (370,277 to 20,528 points) while preserving geometric structure.
- Debugged power, sensor, and I/O reliability issues across autonomous vehicle subsystems, translating hardware constraints into software and systems-level fixes for perception reliability.

Consult Your Community, UT Dallas Jan 2026 – Present
Engineering Lead & Senior Consultant
- Owned Glydr's gaming-hub delivery from requirements to handoff; secured client adoption and reached 1,500+ users.
- Reduced page-load time from 3.9 to 1.4 seconds (64%), API response time from 610 to 220 ms (64%), and interaction delay from 280 to 110 ms (61%) through asset and query optimization, caching, and fewer redundant requests and rerenders.

PROJECTS

Sticky – Full-Stack Task & Reminder Platform
TypeScript, Next.js, React, Hono, Supabase, PostgreSQL, TanStack Query, Zod, Vercel
- Deployed a private task-management PWA with recurring tasks, durable reminders, calendar planning, and realtime sync.
- Kept task writes and reminder events atomic with a transactional outbox; enforced scoped, idempotent commands through a versioned API and owner-level database access policies.

Atlas – Fault-Tolerant Distributed Crawl, Index & Search Platform
Python, FastAPI, PostgreSQL, Redis/RQ, OpenSearch, Docker, AWS, Terraform, React, Vite
- Recovered crawl and indexing jobs after worker failures using persistent scheduling, leases, retries, and outbox delivery.
- Passed a deterministic 10,000-page fault benchmark covering worker loss, complete Redis data loss, retries, duplicate delivery, redirects, malformed content, and transient failures.

JARVIS – Local AI Agent Runtime
Python, React, TypeScript, Rust, Tauri, SQLite, Pydantic, WebSockets, Playwright, Ollama
- Built a local agent runtime connecting durable memory, typed tools, and a desktop interface; enforced action permissions outside the model with explicit approvals, cancellation, and audit records.
- Validated tool execution, undo, replay denial, and backup recovery; passed 396 Python, TypeScript, and Rust tests. Daily-use qualification remains gated on local-model hardware capacity.

WorldState – GPU-Backed Real-Time Vision Runtime
Python, FastAPI, PyTorch, CUDA, OpenCV, WebSockets, PostgreSQL, React, Docker
- Decoupled live vision updates from slow model and storage work using asynchronous inference and shared-memory workers.
- Verified a 30-minute learned-model soak at 30.93 useful updates/s and 33.69 ms p95 with 2,588 MiB flat reserved VRAM, isolating storage latency from the frame-acknowledgement path.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL
Backend & APIs: FastAPI, Django, Hono, Node.js, REST APIs, Server Actions, SQLAlchemy, Alembic, Pydantic, Zod
Data, Search & Infrastructure: PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Supabase Auth/RLS, Docker, Docker Compose, Git, GitHub Actions, Vercel, Doppler, Linux/Unix, CI/CD
AI/ML Systems: BM25, vector search, embeddings, cross-encoder reranking, retrieval evaluation, OpenRouter, Exa, LangChain, PyTorch, NumPy, Pandas, OpenCV
Frontend & Integrations: React, Next.js, TanStack Query, Tailwind CSS, Vite, Framer Motion, Recharts, Stripe, Resend, QStash